Musfikur Rahaman

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PROFESSIONAL SUMMARY

PhD researcher in Information Science with a focus on Explainable AI (XAI), multimodal AI systems, computer vision, and transformer-based deep learning. Published in peer-reviewed venues including Biology Methods and Protocols, IEEE proceedings, and ACM, with ongoing research in hallucination detection, medical vision-language models, and trustworthy AI. Experienced in experimental design, model development, performance evaluation, and scientific writing, with a strong foundation in both foundational and applied AI research spanning healthcare AI, bioinformatics, and video understanding.

EDUCATION

University of Arkansas at Little Rock — Little Rock, AR
Ph.D. in Information Science. — *August 2024 - Present*

Arkansas Tech University
M.S. in Information Technology. GPA: 4.00/4.00. — *Apr 2024*

University of Information Technology & Sciences
B.S. in Information Technology. GPA: 3.83/4.00. — *Apr 2021*

EXPERIENCE

Graduate Assistant — Aug 2024 – Present
School of Counselling, Human Performance & Rehabilitation, University of Arkansas at Little Rock — *Little Rock, AR*

- Conduct research in computer vision and deep learning, with a focus on transformer-based video understanding, Explainable AI (XAI), and trustworthy multimodal AI systems.
- Develop and evaluate AI models for soccer action recognition and medical image report generation, including research on temporal context, hallucination detection, and explanation faithfulness.
- Developed and deployed web-based request and order forms, automating departmental workflows and reducing manual processing overhead
- Led document digitization initiative, converting physical records into structured digital archives to improve data accessibility and compliance
- Supported faculty research on public health disparities, conducting literature reviews and assisting with study design for minority student health studies

Graduate Assistant, — Aug 2022 – May 2024
Division of Computing Sciences, Arkansas Tech University — *Russellville, AR*

- Designed and delivered Data Communication & Networking lab sessions, receiving positive feedback for instructional clarity and engagement
- Served as Teaching Assistant for Software Development and Data Structures courses, providing academic support to 30+ students per semester
- Provided one-on-one mentoring with tailored improvement plans to enhance student understanding and performance
- Presented research at Arkansas Academy of Science, ATU Research Symposium, and LSUS Regional Student Scholars Forum
- Earned 1st Place in Graduate Poster Presentation (Arkansas Academy of Science) and 2nd Place in Graduate Oral Presentation (ATU Research Symposium)

Research Assistant — Jan 2020 – Apr 2021
Department of Information Technology, University of Information Technology & Sciences, Bangladesh — *Dhaka, Bangladesh*

- Co-investigated MLBioIGE, integrating machine learning and bioinformatics to identify genetic factors influencing pulmonary fibrosis in COVID-19 patients, leading to a peer-reviewed publication
- Processed and analyzed genomic and biomedical datasets using statistical techniques, feature extraction, and predictive modeling.

PUBLICATIONS

- MLBioIGE: Integration and interplay of machine learning and bioinformatics to identify the genetic effect of SARS-CoV-2 on idiopathic pulmonary fibrosis patients. Biology Methods and Protocols, May 2022. https://doi.org/10.1093/biomethods/bpac013
- Enhancing the programming sequence for undergraduate computer science students: A program for improving learning outcomes. Proceedings of the 5th HCIRA International Congress, Jun 2023. Vol. 7 Issue 1. DOI: 10.1109/HORA58378.2023.10156793
- Soccer Action Detection Using TimeSformer: A Transformer-Based Approach to Video Understanding. Journal of Computing Sciences in Colleges, 41(10), 72-81, 2026. ACM Digital Library. DOI: 10.5555/3820620.3820634
- Enhancing Disease Detection in South Asian Freshwater Fish Aquaculture Through Convolutional Neural Networks. ATU Scholars Symposium, Apr 2024. https://orc.library.atu.edu/atu_rs/2024/2024/27
- League of Learning: Deep Learning for Soccer Action Video Classification. ATU Scholars Symposium, Apr 2024. https://orc.library.atu.edu/atu_rs/2024/2024/17

PROJECTS

Beyond the Output: Faithful XAI for Medical Image Report Generation
Researcher
- Investigating hallucination detection and explanation faithfulness in medical vision-language models using chest X-ray report generation as a testbed
- Designing a unified evaluation framework linking visual saliency maps, model reasoning chains, and clinical correctness metrics to quantify and reduce unfaithful explanations (in progress)
- Technologies: PyTorch, TensorFlow, Vision Transformers, Medical Imaging, XAI

Image Forgery Detection: ResNet50 + U-Net + Grad-CAM
Researcher
- Built an explainable computer vision pipeline for detecting and localizing image manipulation, achieving AUC 0.86 and recall 0.88
- Integrated Grad-CAM to produce pixel-level interpretability maps, making predictions auditable for downstream forensic use
- Technologies: ResNet50, U-Net, Grad-CAM, PyTorch, Computer Vision, XAI

NewsLogic: AI-Powered Fake News Detection System
Researcher
- Designed and deployed a full-stack AI system combining BERT, DistilBERT, and rule-based validation for multi-signal fake news classification
- Built FastAPI backend with real-time explanation generation using LLaMA-based models, enabling transparent inference for end users
- Technologies: BERT, DistilBERT, LLaMA, FastAPI, Python, NLP

SKILLS

- **Machine Learning & AI Research:** PyTorch, TensorFlow, Keras, HuggingFace Transformers, Scikit-learn, ViT, TimeSformer, BERT, DistilBERT, LLaMA
- **Explainability & Vision:** Grad-CAM, XAI, Computer Vision, Medical Imaging, ResNet50, U-Net
- **Programming & Data:** Python, SQL, C++, Java, NumPy, Pandas, Matplotlib, Jupyter Notebook
- **APIs, Deployment & Databases:** FastAPI, REST APIs, Git, GitHub, GitLab, MySQL, MongoDB, PostgreSQL, SQL Server, Supabase, Render, VS Code

AWARDS & HONORS

- 2nd Place, Graduate Oral Presentation — ATU Research Symposium, Jan 2024. Project: League of Learning: Deep Learning for Soccer Action Video Classification
- 1st Place, Graduate Poster Presentation — Arkansas Academy of Science, Jan 2023. Project: League of Learning: Deep Learning for Soccer Action Video Classification